Hochschule
Bielefeld

University of
Applied Sciences
and Arts

AIR Project 2025

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20.06.2025

Center for
Applied Data Science
Gütersloh

PREVIEW
Agenda Today

- Safety instructions
- Lab-Tour
- Introduction: Robotino & Robotnio Factory
- Communication towards Robotino: Direct/Master
- Project Task

20.06.2025

SAFETY INSTRUCTIONS.

HSBI

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Safety Instruction

Specific Part
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SAFETY: WHAT & WHY

- Work together in a safe, responsible and healthy way.
- Know what actions to take in the case of an emergency

All safety instructions must be followed without exception!

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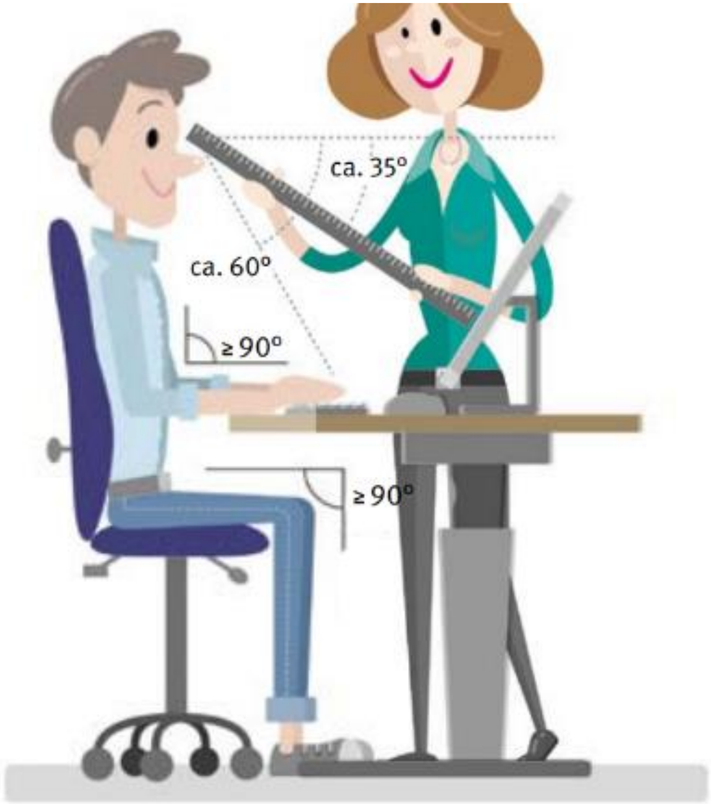
HEALTHY WORKING AT YOUR COMPUTER

- Work together in a safe, responsible and healthy way.
- Know what actions to take in the case of an emergency

All safety instructions must be followed without exception!



German Guide:
https://www.vbg.de/SharedDocs/Medien-Center/DE/Broschuere/Themen/Bildschirm_und_Bueroarbeit/Gesund_arbeiten_am_PC_Faltblatt.html

BE AWARE OF FURTHER GUIDANCE IN THE LAB

- Allgemeine Labordnung
  - General rules how to behave in the Laboratory
- Betriebsanweisungen
  - Essential information on risks and how to deal with them per machine type
  - Read before use!
- Stickers and writings on the machines

Use translation apps and ask questions if something is unclear!

Shape	Base Meaning	Examples
	Prohibition	- No smoking - Do not Touch
	Command	- Wear PPE - Wash Hands
	Warning	- Hot surface - Electric danger - Opposing roles
	Safety	- Emergency exit - First aid
	Fire protection	- Fire alarm - Fire extinguisher

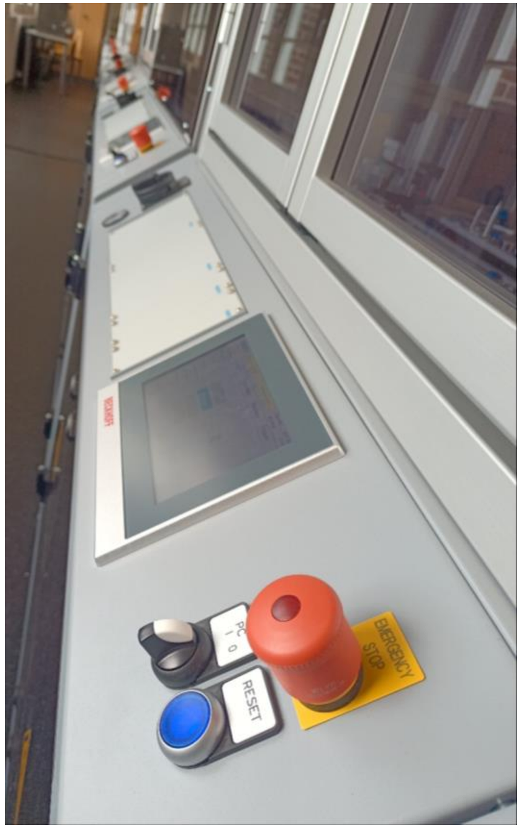
Reference DIN EN ISO 7010: http://www.iso7010.de/ ASR A1.3

EMERGENCY STOP / ELECTRIC DISCONNECT

- No Emergency-Off in the room!
  - Cause: robots actively break

- Using Emergency Stop
  - Physically connected modules will immediately stop
  - Buttons at the exits have NO EFFECT!

- Cut Power: in an electric emergency there are black main-switches that kill the power for each module!



Work with electricity

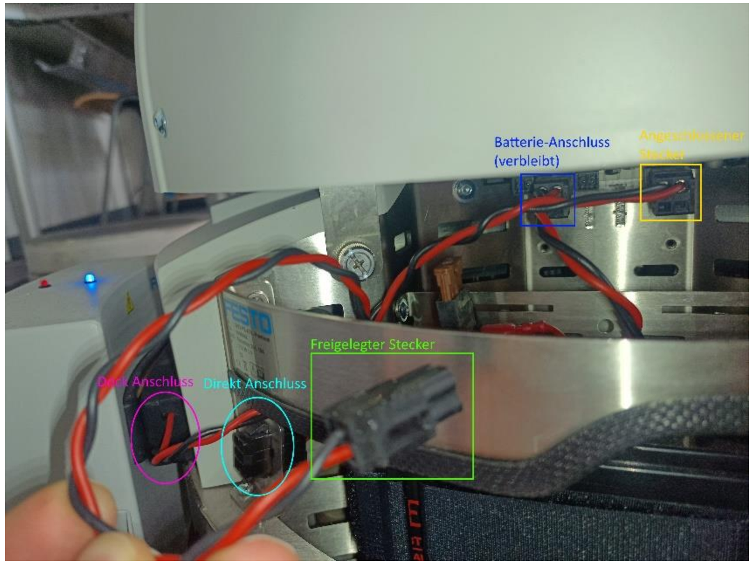
No electrical work in the factory by any students!

Only work in a voltage-free environment!

Switch on only after review by supervisor!

5 safety steps to consider:

1. Flip the breaker / pull the cord
2. Secure against being switched on again
3. Doublecheck if voltage-free
4. Earth/Shortcircut
5. Cover or shut off nearby live parts



See original source.: https://fh-bielefeld.agu-hochschulen.de/ablauforganisation/unterstuetzende-prozesse/labor-und-werkstatt/werkstatttaetigkeiten/experimentiertaetigkeiten-mit-elektrischem-strom

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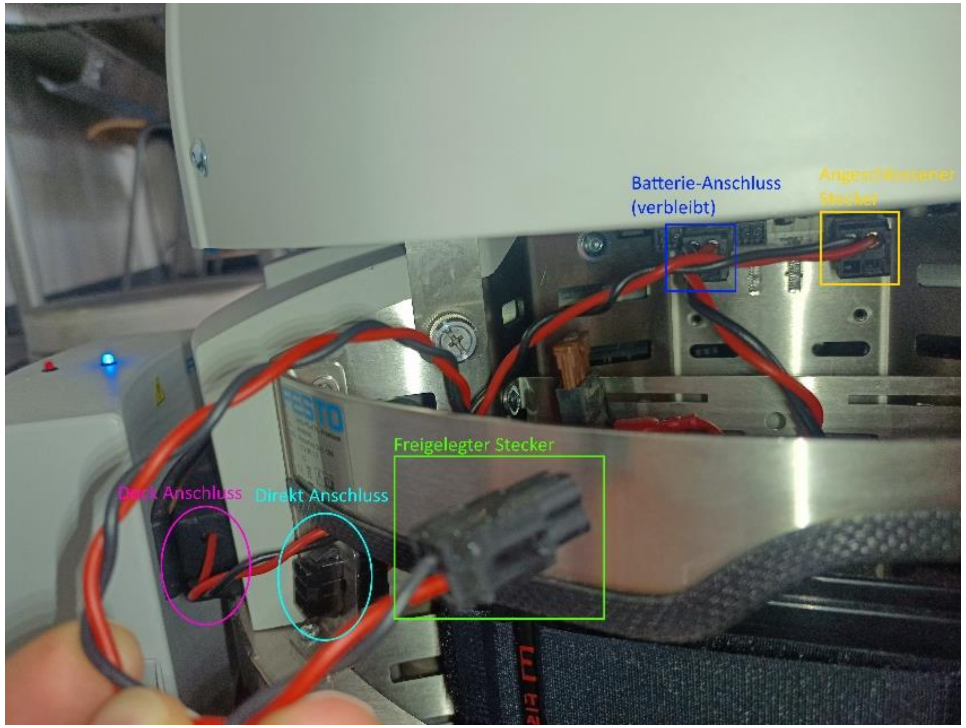

Vgl.: https://hsbi.agu-hochschulen.de/ablauforganisation/unterstuetzende-prozesse/labor-und-werkstatt/werkstatttaetigkeiten/experimentiertaetigkeiten-mit-elektrischem-strom

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Safety Instruction

General Part
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LAB-TOUR.

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Mandock

Station for feeding parts into the system (delivery) and issuing them from the system (shipping)

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Box-Storage ASRS-B

Fully automated high-bay storage system that stores boxes that contain mixed parts. It is used by Robotino AGVs.

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Pick&Sort

Commissioning of supply-parts: Preparing box content based on needs of other factory modules.

Palletizing Robot PalRob-B

Transfer parts between boxes and carriers

Pallet storage ASRS-P

Inline high bay storage for pallets (contains either empty pallets, back covers or finished products)

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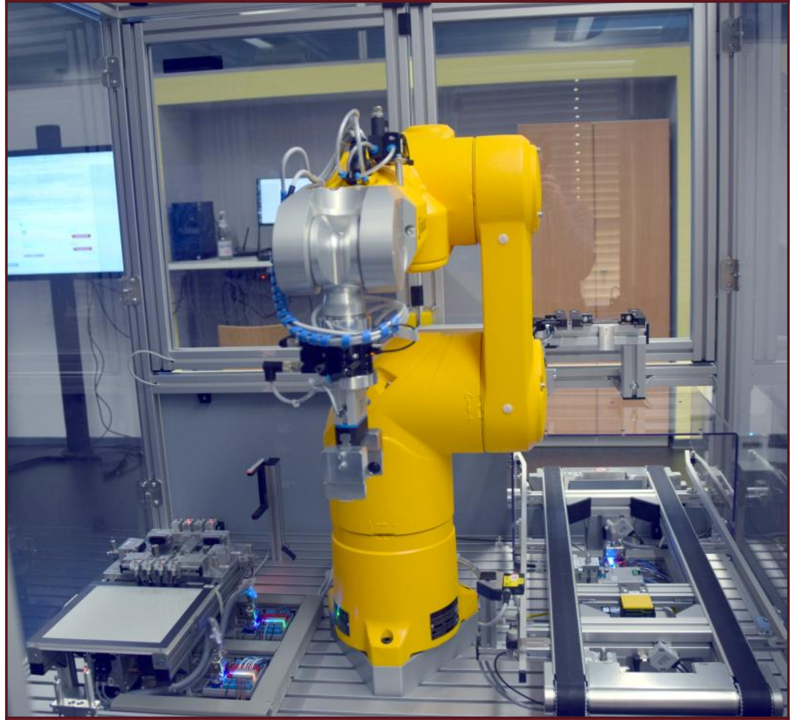
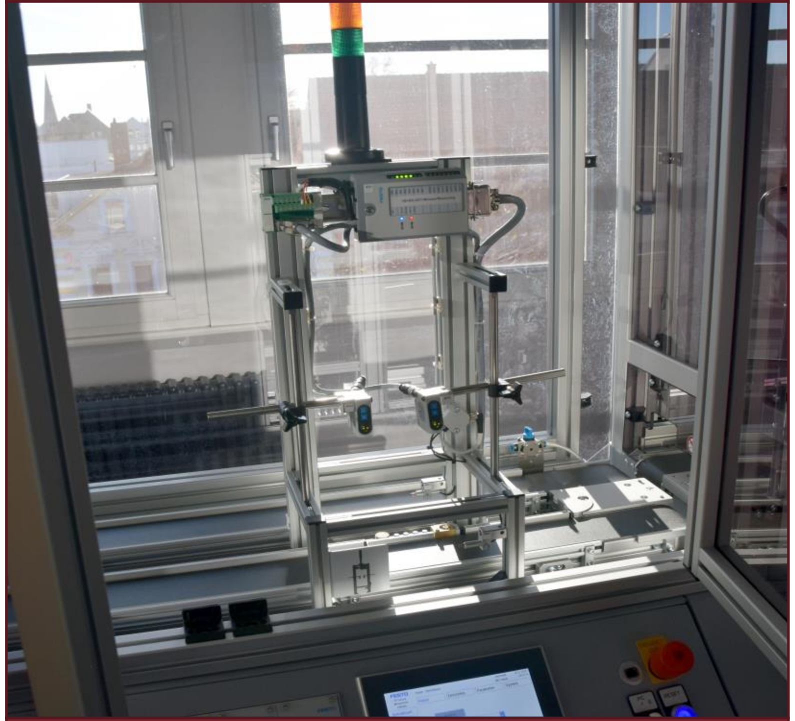
Robot Assembly Station RASS1



(Dis-)Assembles the back cover, mainboard and display. During the process the workpiece is turned on (/off) and a wireless connection is established.

Analogue measuring station

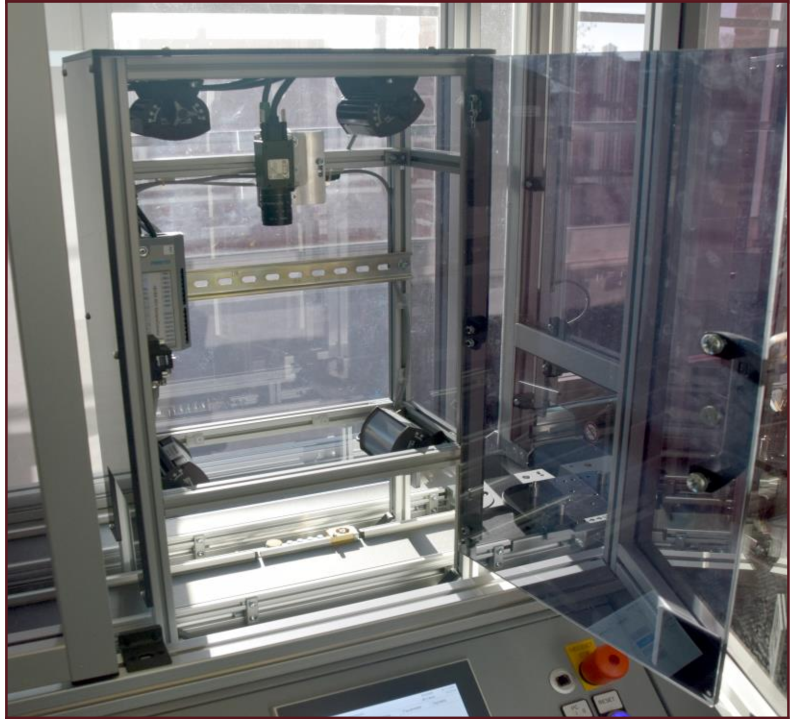


Checks hight tolerances with laser based distance sensors.

Robot Assembly Station RASS2

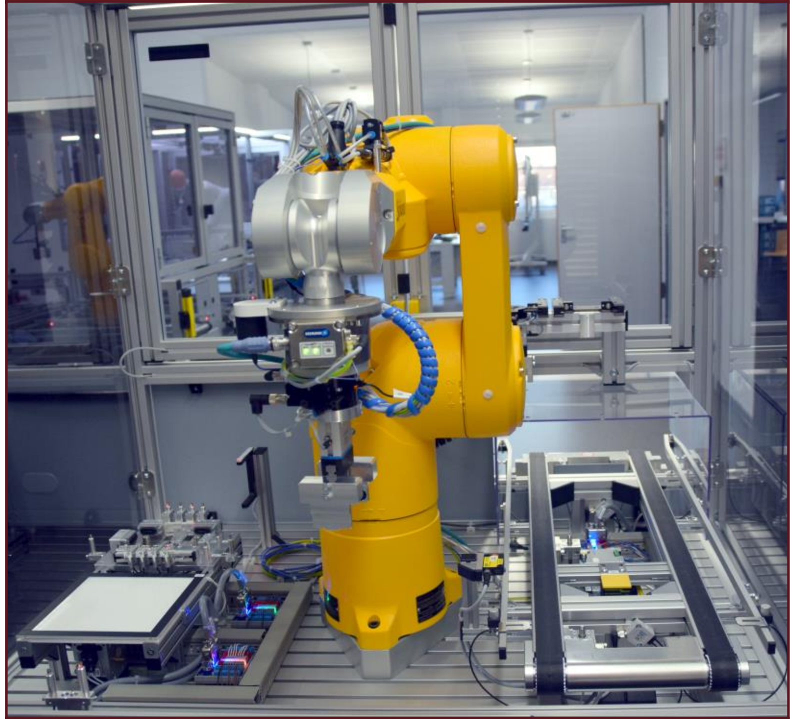
(Dis-)Assembles the sensor boards based on the custumers order.

Camera inspection



Tests if all ordered shields (display, gyroscope, weather,...) are mounted on the correct position by reading their barcodes.

Robot Assembly Station RASS3



Assembly (and disassembly) of front covers.

Funktionstest

Quality check after completed assembly. (Tests: touch on touchscreen, read/write of analogue shield, check values of gyroscope and weather sensor, wireless connectivity)

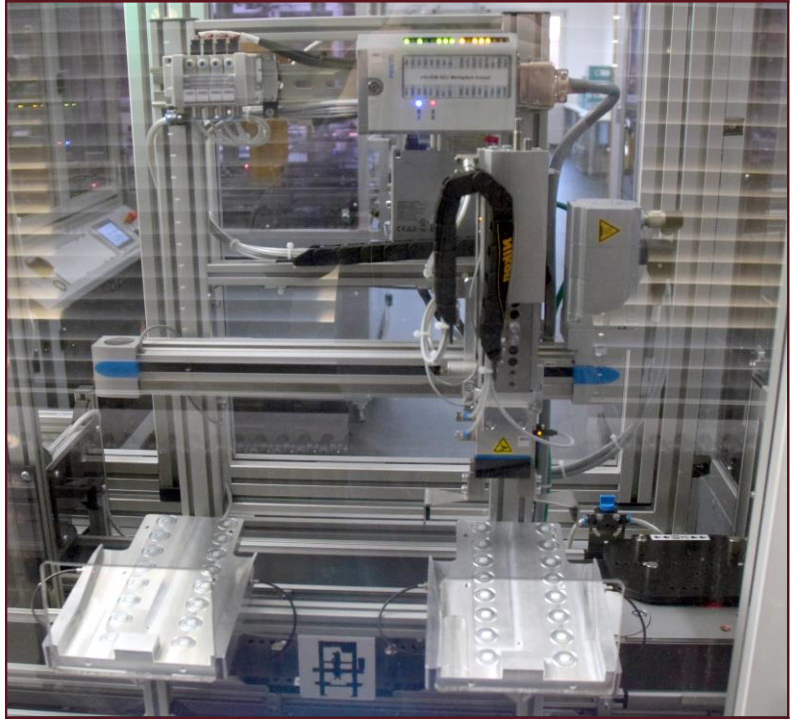
Switch

Diverts empty carriers for potential pallet loading and directs finished workpieces to the label printer.

Labeling station

Prints and applies labels to finished products, including the current date and a unique QR code.

Manual output robot



Up to four finished products can be handed out directly instead of forwarding them to storage or shipping.

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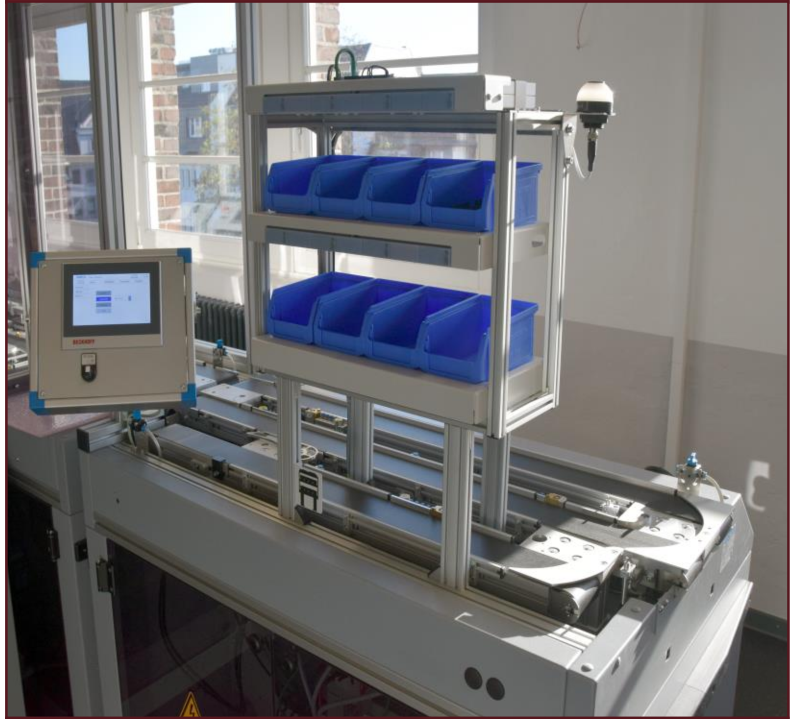
Manual Rework



Workers can do rework if problems were found during the automated tests.
User is supported by a pick-by-light system.

3D-Printing

Printing customized frontcovers based on customer order. Faild prints can be recycled into new printing filament.

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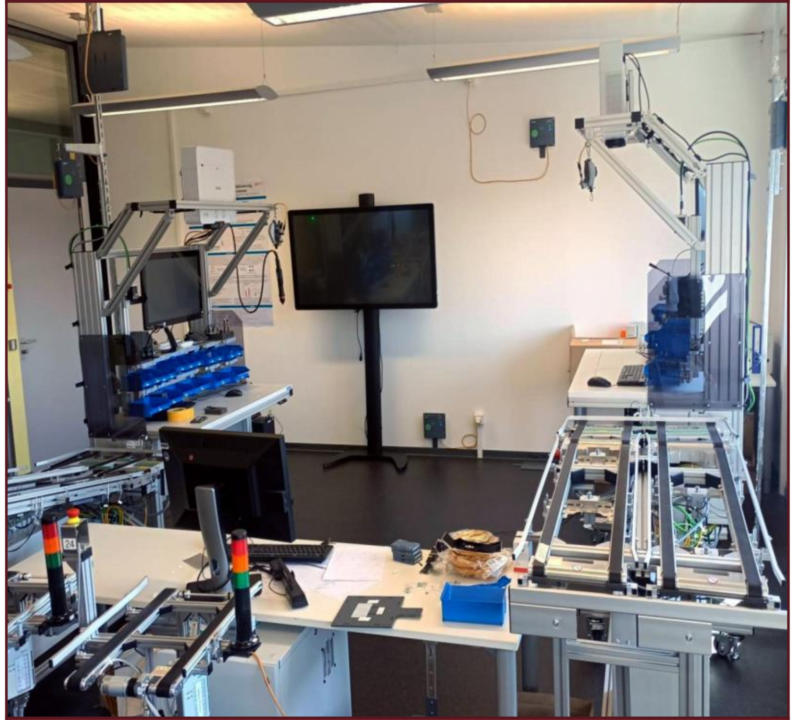
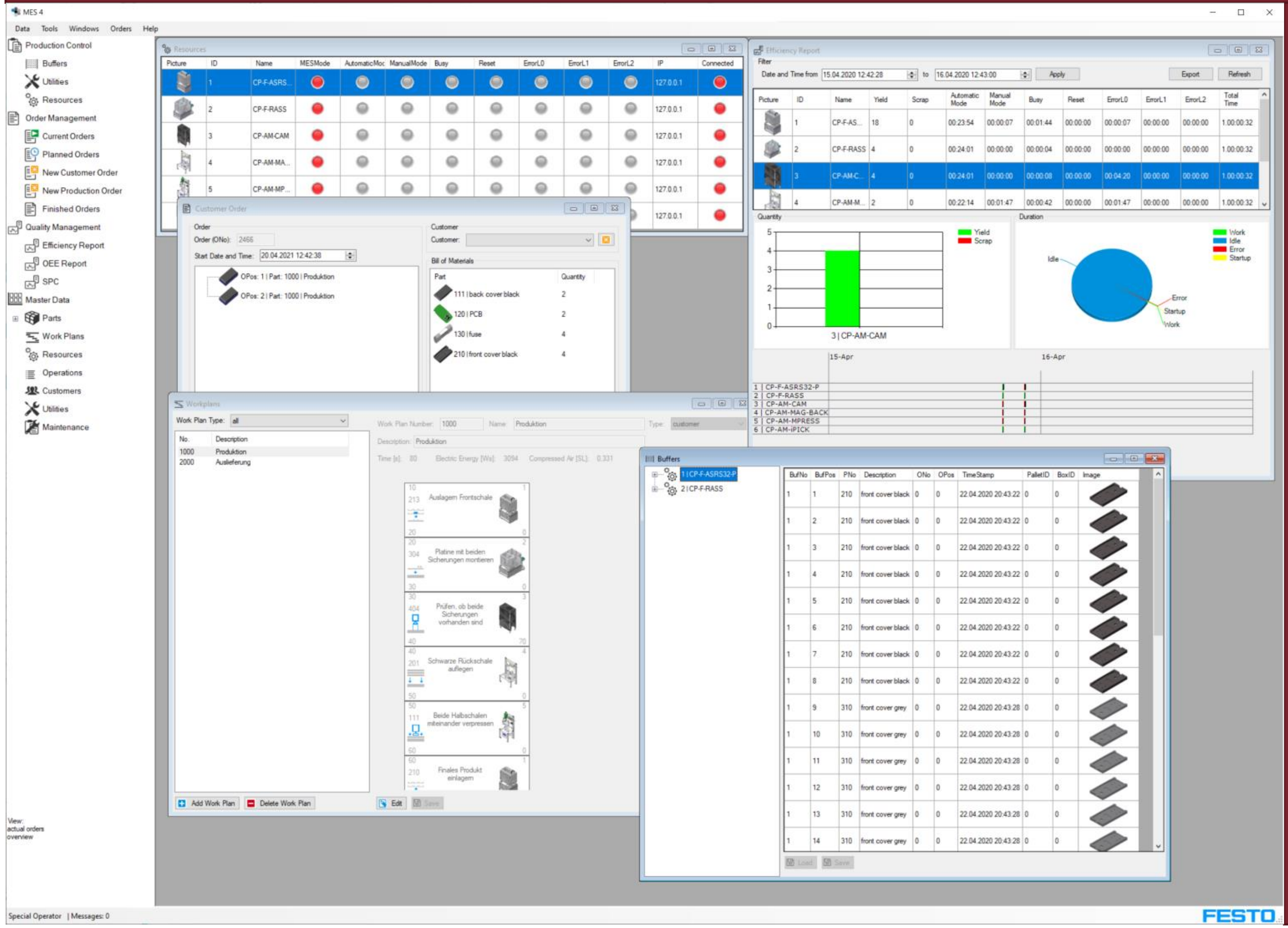
Assisted Workplaces



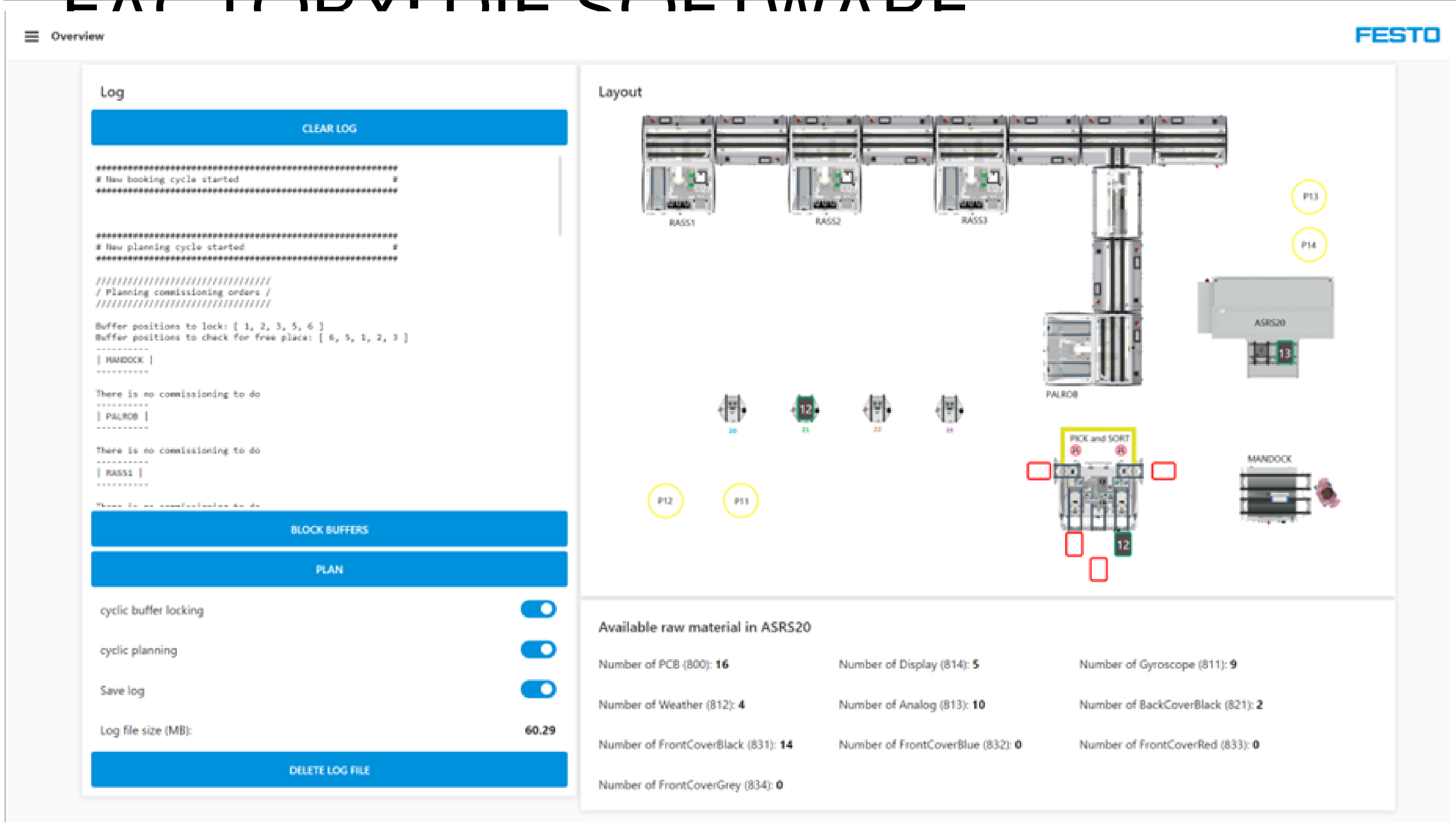
Assisted assembly of complex or highly customized components by personnel.

MES: Manufacturing Execution System



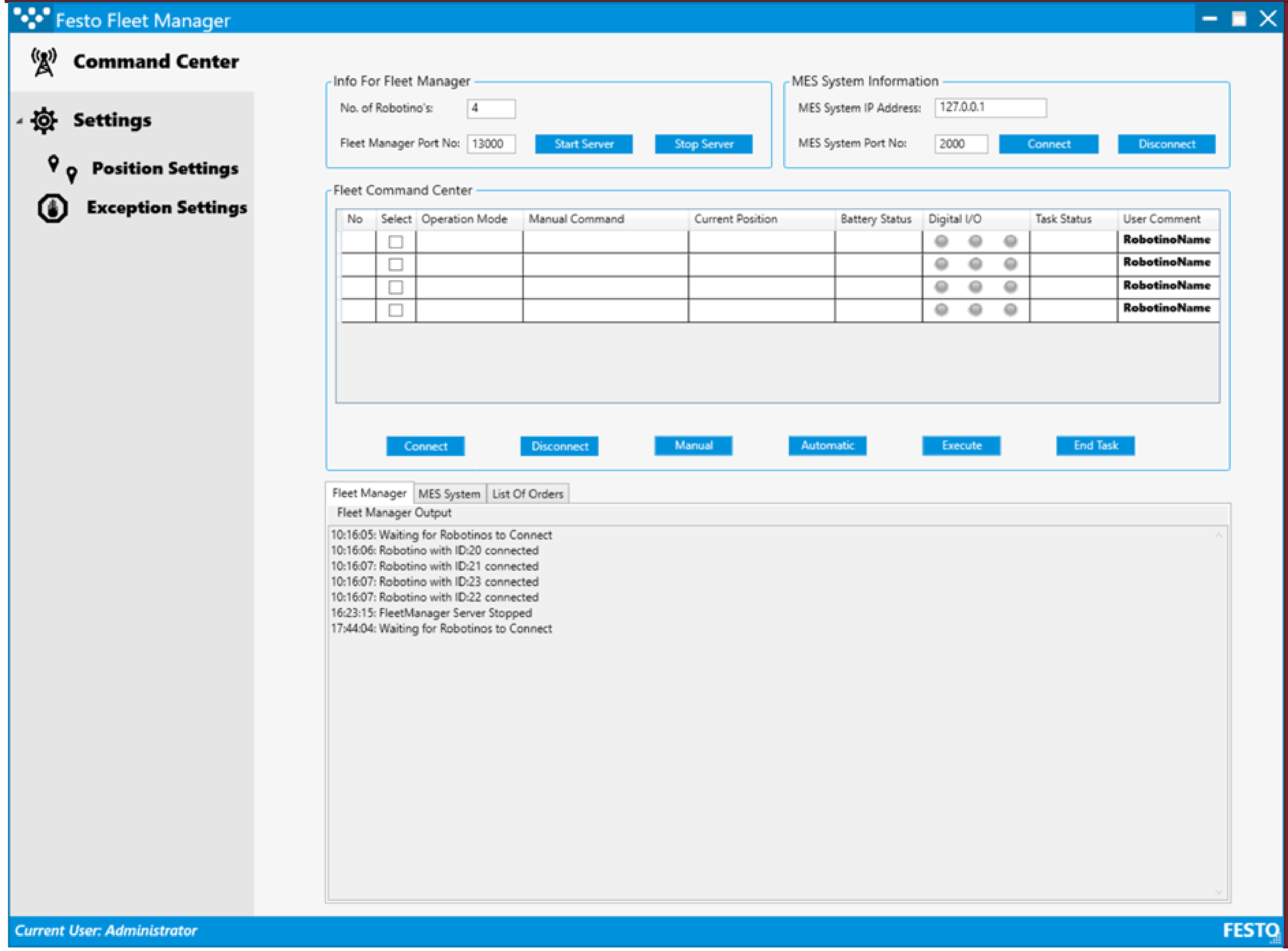
- User interface for managing the production system
- Order execution
- Communication with the IPC's
- Workplans
- Product configurations
- Ressource state

Logistics Planner (Deprecated – HSBI Optiflow soon)



- User interface visualizing storage contents and the state of parts-deliveries in the factory
- Organizing material supply

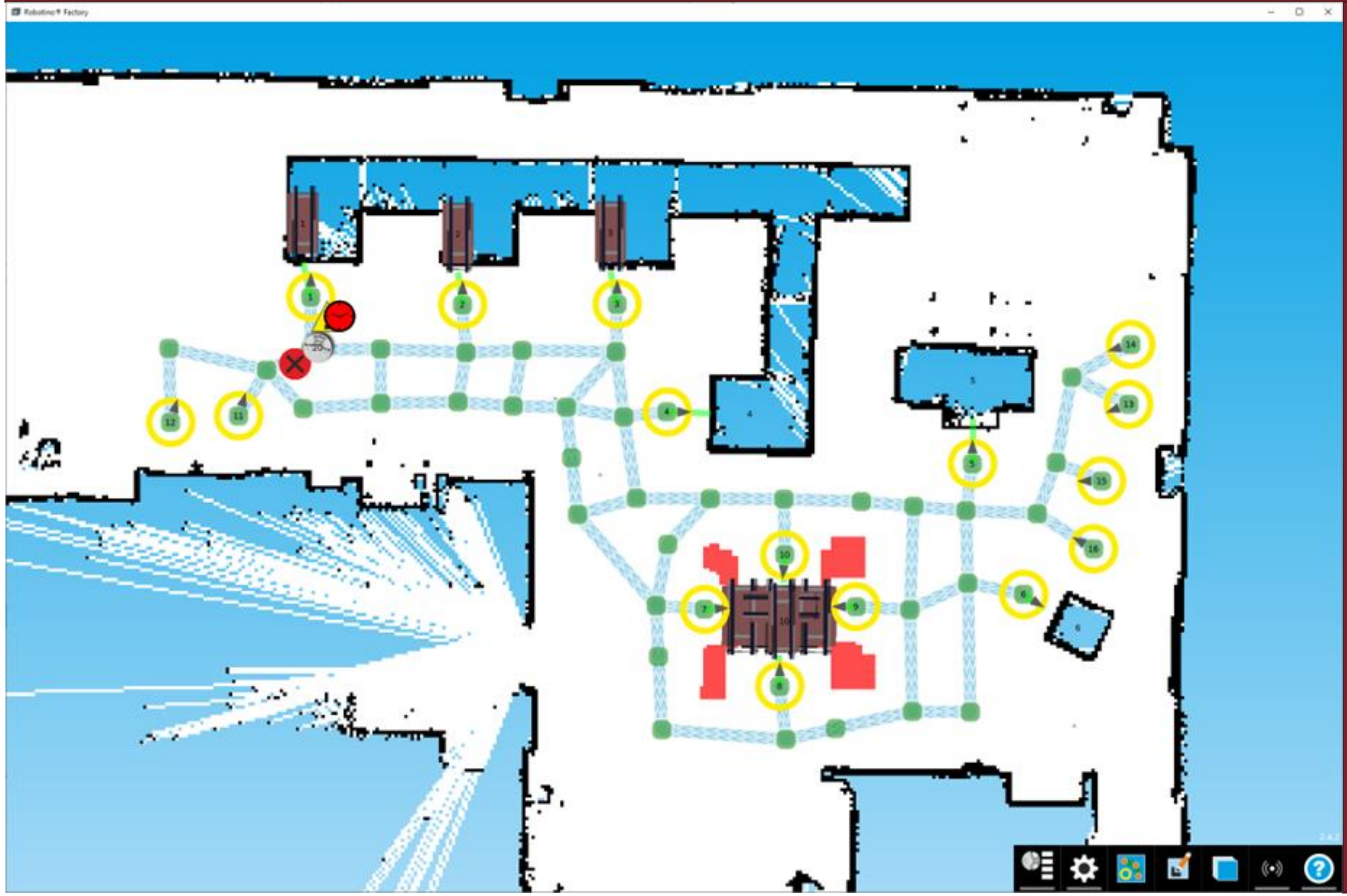
Fleetmanager (Deprecated – HSBI Optiflow soon)



AGV management software for fleet initialization, charging control, job assignment*, and error resolution.

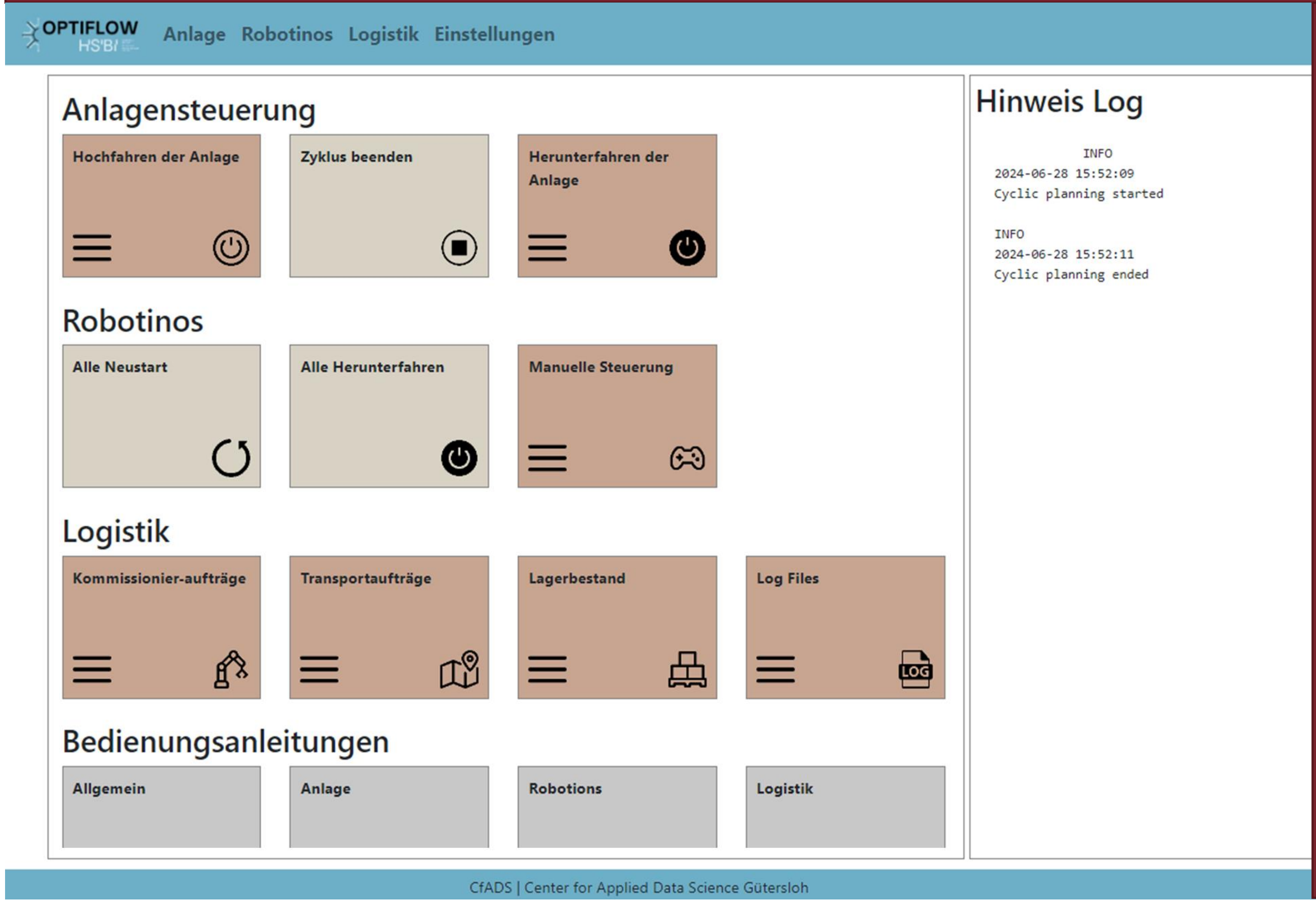
* based on position, Robotino state, current tasks, and MES data

Robotino Factory



- Initializing and visualizing the current Fleetstate of all AGV
- Interface to edit the navigation Map
- Interface to manually control the Robotino-AGVs

HSBI-Optiflow



Upcoming software solution by HSBI students to manage the factory and organize material supply.

IPC Programs

FACTORY: DIE SOFTWARE

Quelle: https://infosys.beckhoff.com

Each module has its own industrial computer that manages all processes on the station e.g.:

- Belts
- Stoppers
- Robot arm
- Belt switches
- Sensors

Extends hardware safety by software safety and operates based on MES instructions (e.g. “mount display”) and provides access to lots of data points.

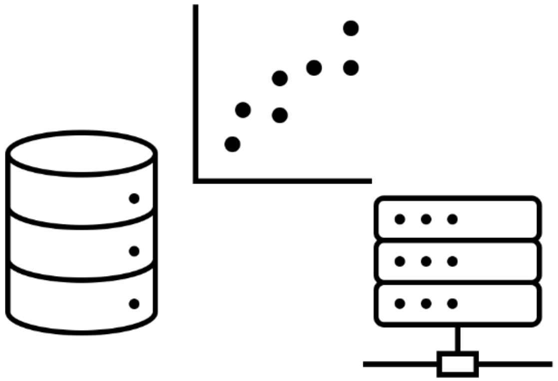
Mostly Beckhoff, some Siemens

SmartProduct Firmware

- accessible web interface
- Hotplug of various sensor-shields
- Provides interfaces (REST, Websocket) for effective wireless communication

Data analysis scipts

- Preprocessing, storing, replicating, machine-learning, AI-applications etc.
- BigData-driven applications running on DA-Cluster of CfADS for data science research in multiple fields like:
    - Degradation Analysis
    - ChangePoint Detection
    - Scheduling



ROBOTINOS.

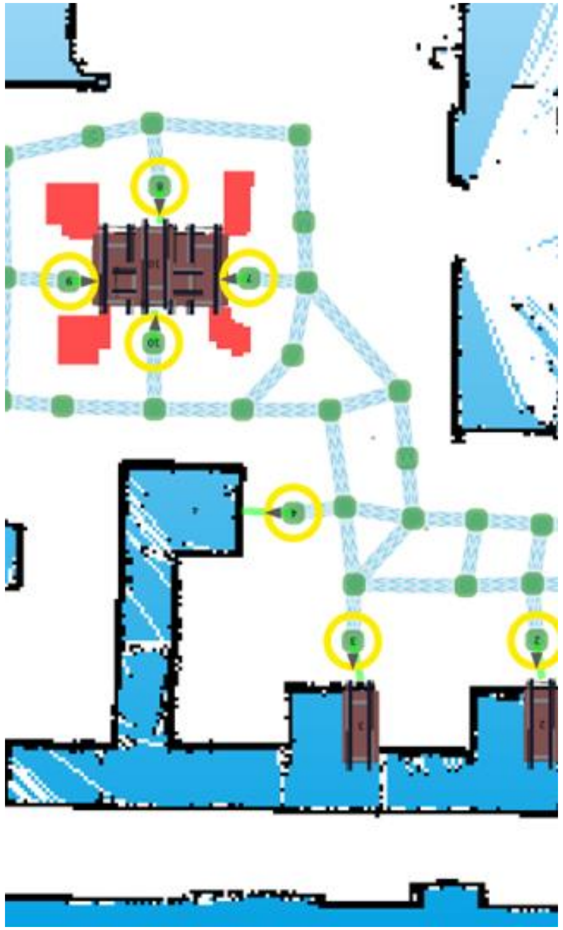
ROBOTINO AGV – FLEXIBLE MOBILE ROBOT

- **What is Robotino?**
  - **Mobile robot platform** for education, research, and industrial automation.
  - **Omnidirectional movement** (Mecanum wheels) for flexible navigation.
  - **Autonomous or remote-controlled** operation.
  - Used in **material transport** between stations via conveyor belt add-on.
- **Sensor Overview**
  - Robotino is equipped with multiple sensors for autonomy:
  - **Laser distance sensor** (e.g., SICK LMS100) for obstacle detection.
  - **Optical encoders** for precise wheel movement tracking.
  - **Infrared sensors** for proximity detection.
  - **Camera** for vision-based navigation.

ROBOTINO FACTORY: PATH PLANNING & CONTROL

- **What is Robotino Factory?**
  - **Graphical programming software** by Festo for AGV automation.
  - Used to **teach paths, define workflows**, and simulate tasks.
  - Compatible with **Robotino fleet management** (multi-AGV control).

- **Path Teaching & Network Creation**
  - **Waypoint Recording:**
    - Drive Robotino manually (via joystick) to **record path nodes.**
    - Adjust parameters (speed, pauses, orientation) in the GUI.
  - **Path Editing:**
    - Connect nodes to form **routes**

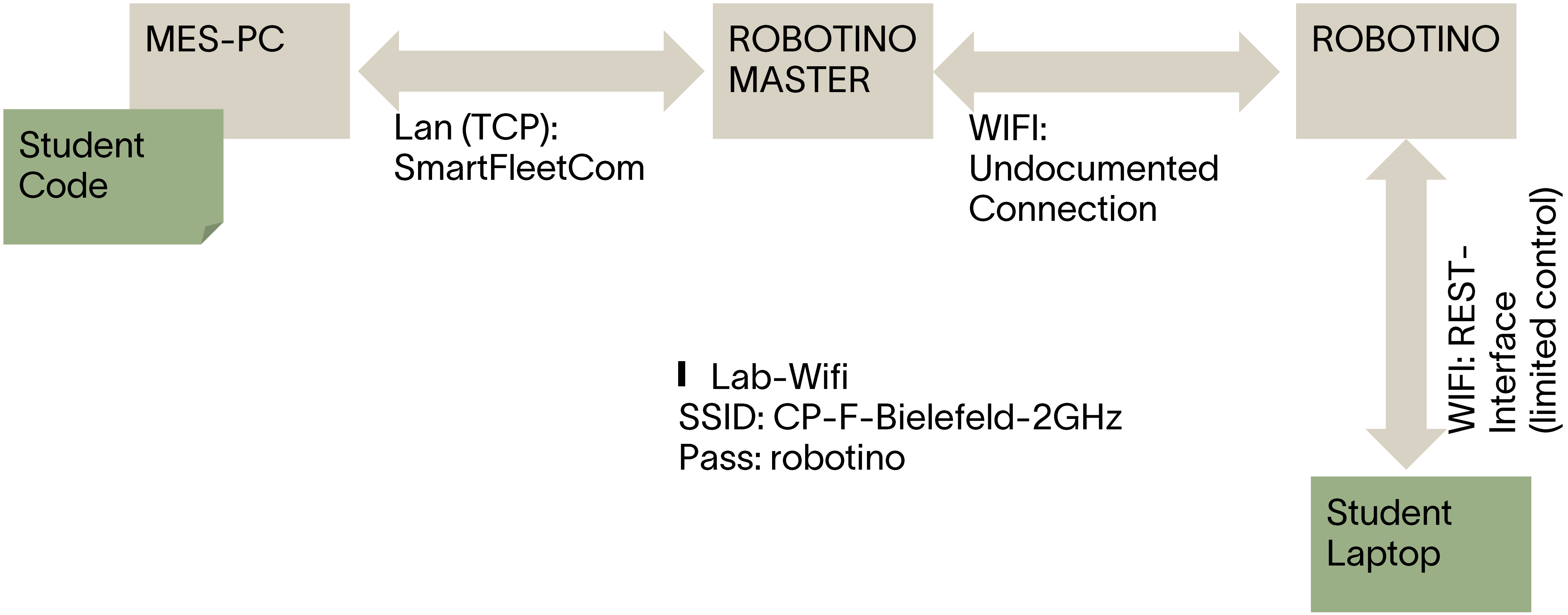


ROBOTINO FACTORY: PATH PLANNING & CONTROL

- Tools and options
  - Station Types:
    - **Parking/Charging**: Idle zones with timeout/battery control.
    - **Load/Unload**: I/O-triggered (conveyor/PLC).
  - Path Types
    - **Uni-/Bidirectional**: Adjust collision zones.
    - **Express Routes**: High-speed priority paths.
    - **Width Settings**: Narrow or wide for obstacle avoidance

"Today, creating path networks is 100% manual work – slow, assumption-driven, and dependent on expert knowledge. We lack tools to automatically learn or adapt to new factory layouts efficiently."

ROBOTINO: COMMUNICATION



- Lab-Wifi
SSID: CP-F-Bielefeld-2GHz
Pass: robotino

Preparation Task

- Retrieve Sensor Data of the Robotino
- Move the robotino by connecting to the Master

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Project Task

- Record a map including stations
- Generate an optimized Path-Network based on the recodings

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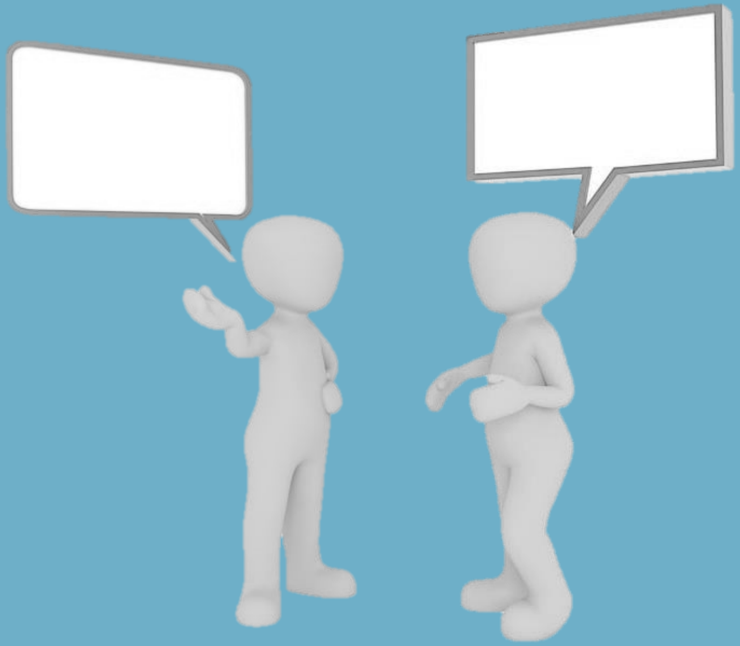

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